Safety Helmet Detection Using YOLOv8

AIRI Team PITB - AI Internship Task 1
End-to-End Computer Vision Object Detection Project

Intern:	Hamza Nadeem
Submitted to:	Omar Farooq - Associate AI/ML Engineer

Date:	17 October 2026
Model:	yolov8n.pt (best of 2 experiments)
Platform:	Google Colab - Tesla T4
Test mAP@0.5 / mAP@0.5:0.95:	0.679 / 0.401

1. Problem Definition

This project detects safety helmets and no-helmet cases in workplace images and video. The system supports safety monitoring by identifying people who are not wearing helmets and flagging the scene as a violation.

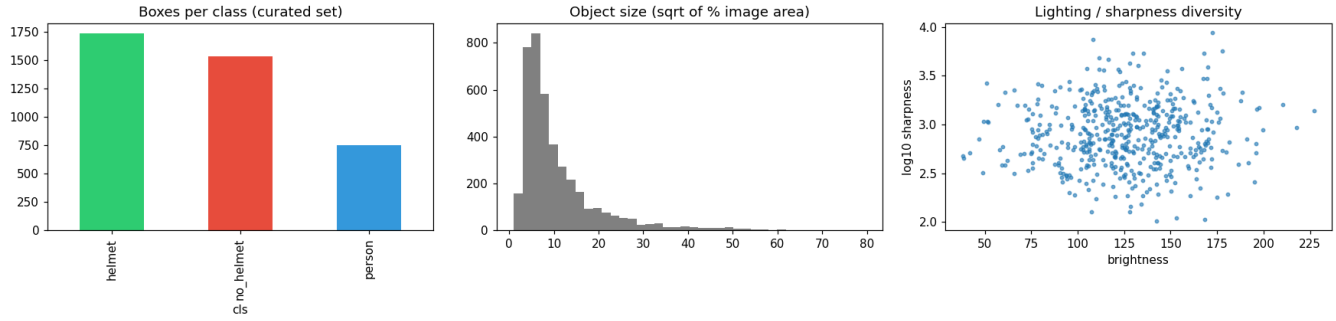
Question	Answer
What is detected?	Heads with a helmet, bare heads (no helmet) and people
Why is it useful?	Head injuries are severe and helmet rules are hard to enforce manually; automatic detection flags violations continuously.
Where can it be applied?	Construction sites, factories, warehouses, mines, plants, motorcycle road safety - anywhere CCTV exists.
Target classes	0: helmet, 1: no_helmet, 2: person
Final output	Boxes + labels + confidence scores, and a COMPLIANT / VIOLATION status per image or frame.

2. Data Collection

Source: Kaggle: andrewmvd/hard-hat-detection (Hard Hat Workers, MakeML) - CC0 public domain. Original MakeML Hard Hat Workers dataset, also mirrored on Roboflow Universe. The CC0 license permits use and publication. Each image's origin is recorded in outputs/data_analysis/dataset_sources.csv.

- Raw images with annotations: 5000
- Near-duplicates removed with perceptual hashing (Hamming <= 5): 92
- Degenerate / tiny boxes removed: 11; duplicate boxes removed: 0
- Curated final set: 500 images (diversity-aware selection guaranteeing all classes and difficulty conditions)

Statistic	Value
images selected	500
boxes: helmet	1737
boxes: no_helmet	1533
boxes: person	751
images containing helmet	380
images containing no_helmet	270
images containing person	158
images tagged small	233
images tagged dark	78
images tagged blurry	100
images tagged crowded	225
images tagged easy	119



Class balance, object sizes and lighting/sharpness diversity of the curated set



Easy and difficult examples (small, dark, blurry, crowded)

3. Data Annotation

Tool: Roboflow Annotate, exported in YOLOv8 format. Manual Roboflow batch of 120 images prepared (import pending). Guidelines: tight boxes, every visible object labelled, one box per object, minimal background, unclear objects not guessed, partially visible objects only when the class is clear, consistent class names (no_helmet = bare head incl. caps/hoods/hair), and a second review pass.



Final YOLO labels read back from disk (green helmet, red no_helmet, blue person)

4. Dataset Organisation and data.yaml

split	images	manual_images	helmet	no_helmet	person	% of images
train	349	0	1183	1034	525	69.800
val	100	0	326	326	127	20.000
test	51	0	228	173	99	10.200

Labels use the YOLO format <class_id> <x_center> <y_center> <width> <height>, normalised to 0-1. The split is stratified by class content, difficulty and annotation origin, and every image belongs to exactly one split.

```
path: /content/drive/MyDrive/cv_project/dataset
```

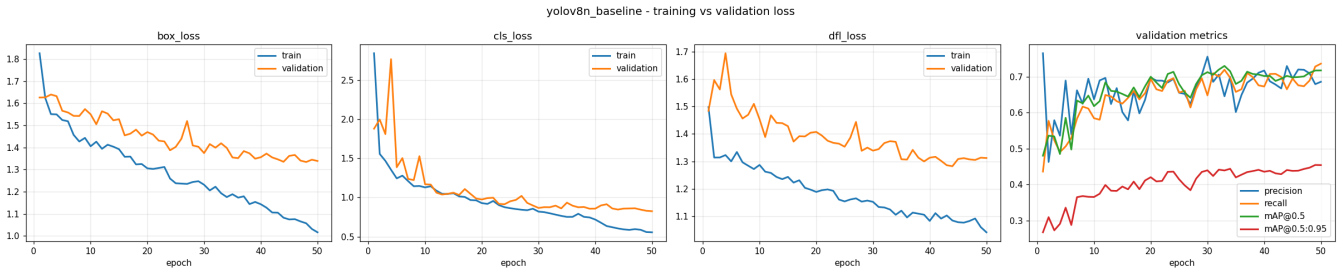
```
train: images/train
val: images/val
test: images/test
names:
  0: helmet
  1: no_helmet
  2: person
```

- path: dataset root; other paths are relative to it
- train: training images (labels found by replacing images/ with labels/)
- val: validation images, evaluated every epoch to pick best.pt
- test: held-out images used once for final results
- names: class id -> class name mapping used in the label files

5. Model Training

name	weights	epochs	imgsz	batch	optimizer	train_minutes	gpu	best_epoch	val_mAP50	val_mAP50_95
yolov8n_baseline	yolov8n.pt	50	640	16	AdamW	6.900	Tesla T4	49	0.717	0.455
yolov8s_exp2	yolov8s.pt	50	640	16	AdamW	6.500	Tesla T4	47	0.722	0.451

Pretrained COCO weights, image size 640, seed 42, no early stopping. Best model on validation: yolov8n_baseline (copied to models/best.pt).



Training vs validation loss (box, classification, DFL) and validation metrics

6. Evaluation

split	precision	recall	mAP50	mAP50_95
val	0.679	0.726	0.717	0.454
test	0.727	0.655	0.679	0.401

Per-class performance

split	cls	instances	precision	recall	mAP50	mAP50_95
val	helmet	326	0.832	0.864	0.910	0.588
val	no_helmet	326	0.815	0.896	0.915	0.591
val	person	127	0.390	0.417	0.326	0.182
test	helmet	228	0.850	0.793	0.856	0.504
test	no_helmet	173	0.828	0.919	0.921	0.577
test	person	99	0.503	0.253	0.259	0.122

- Precision = TP/(TP+FP): of all predicted boxes, the share that is correct (few false alarms).
- Recall = TP/(TP+FN): of all real objects, the share the model finds (few misses).
- mAP@0.5: mean over classes of the area under the precision-recall curve, a hit needs IoU >= 0.5.
- mAP@0.5:0.95: the same averaged over IoU 0.50-0.95, rewarding precisely placed boxes.

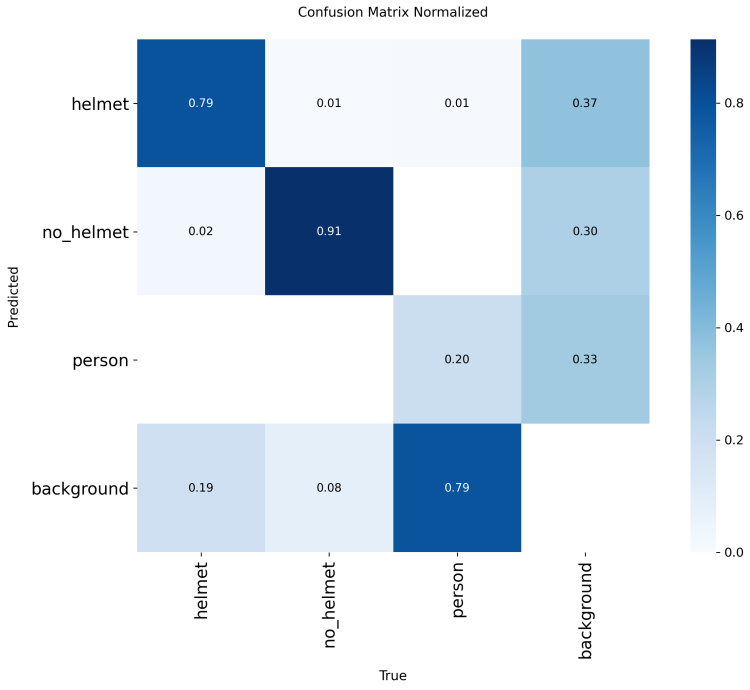
Best class: no_helmet - test mAP@0.5:0.95 = 0.577, mAP@0.5 = 0.922

(1034 training instances; median box = 0.39% of the image).

Worst class: person - test mAP@0.5:0.95 = 0.122, mAP@0.5 = 0.259
(525 training instances; median box = 4.04% of the image).

Training instances per class: helmet: 1183, no_helmet: 1034, person: 525

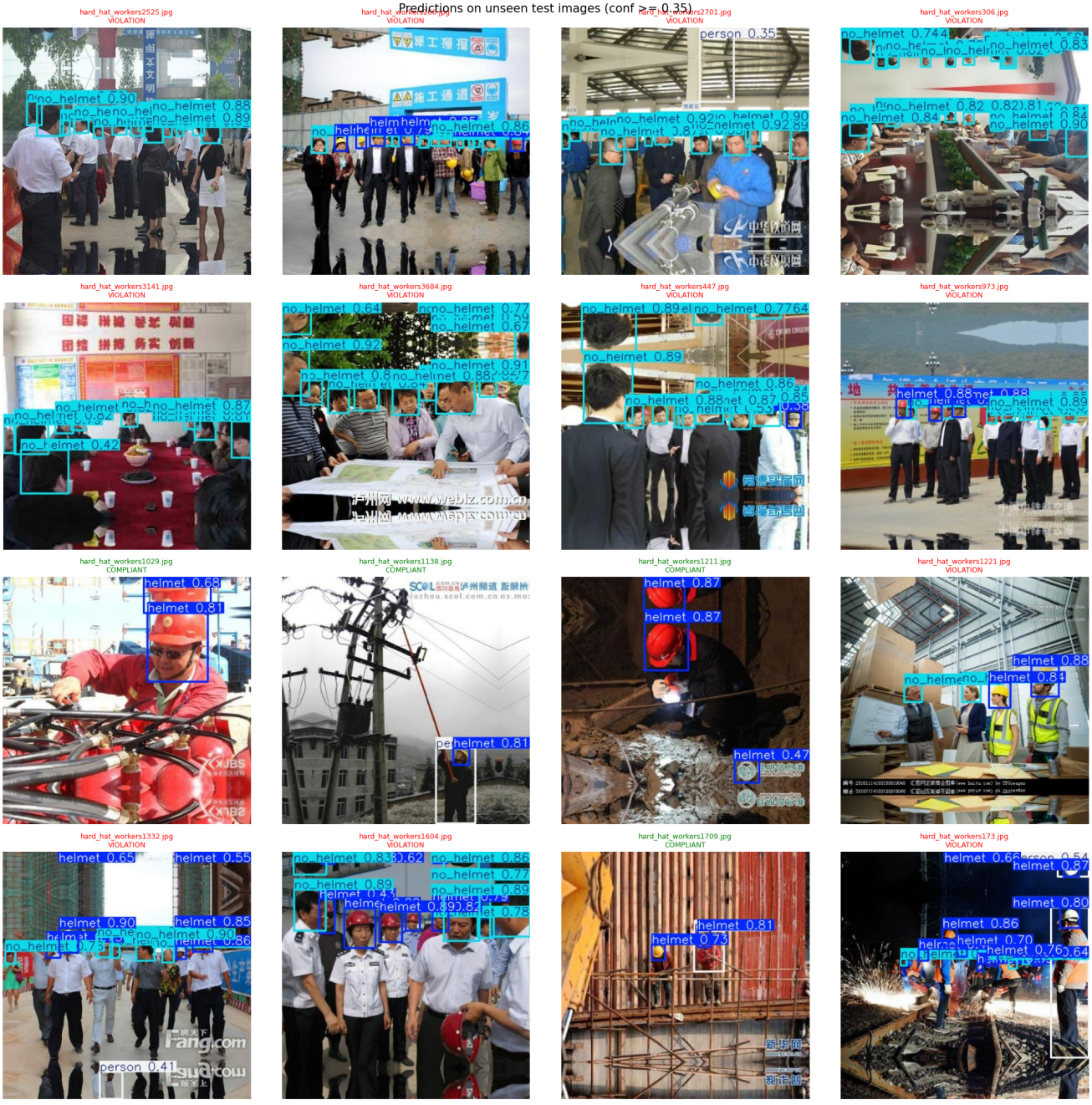
Why the model makes mistakes (evidence from this run):
- Class imbalance - helmet has 1183 training boxes vs 525 for person, so the model sees far fewer examples of the rarer class.
- Small objects - heads and helmets in crowd / far-away shots cover a tiny fraction of the image (median helmet box ~ 0.51% of the image), leaving few pixels to tell a helmet from hair or a cap.
- Look-alikes - caps, hoods, hair and headscarves resemble helmets at low resolution, causing no_helmet <-> helmet swaps.
- Inconsistent person labels in the public source data - many people in the original dataset were never labelled as person, so a correct person detection is sometimes scored as a false positive. The manual annotation batch fixes this for the hand-labelled images.
- Lighting / blur - dark and motion-blurred images reduce contrast at object edges (see error analysis, Phase 10).



Normalised confusion matrix on the test set (conf >= 0.35)

7. Inference on Unseen Data

51 unseen test images predicted at confidence >= 0.35. Status counts: VIOLATION: 31, COMPLIANT: 20. Video: 300 frames processed at 55.8 FPS.



Predictions with bounding boxes, labels and confidence scores

8. Error Analysis

False negative	False positive	Poor bounding box	Wrong class
140	78	7	6

#	Image	Actual Object	Model Prediction	Error Type	Possible Reason
1	hard_hat_workers1029.jpg	person	not detected	False negative	Occlusion / crowded scene
2	hard_hat_workers4352.jpg	helmet	not detected	False negative	Poor lighting; Partially visible at image edge
3	hard_hat_workers1604.jpg	no_helmet	not detected	False negative	Partially visible at image edge
4	hard_hat_workers3408.jpg	person	not detected	False negative	Partially visible at image edge
5	hard_hat_workers3110.jpg	nothing (background)	helmet (0.83)	False positive	Poor lighting; Partially visible at image edge; Likely missing label (annotation mistake)
6	hard_hat_workers2080.jpg	nothing (background)	person (0.78)	False positive	Likely missing label (annotation mistake); Inconsistent person labels in source data
7	hard_hat_workers306.jpg	nothing (background)	no_helmet (0.73)	False positive	Small object; Likely missing label (annotation mistake)
8	hard_hat_workers43.jpg	helmet	no_helmet (0.82)	Wrong class	Similar appearance (cap/hair/hood vs helmet)
9	hard_hat_workers3593.jpg	person	helmet (0.49)	Wrong class	Occlusion / crowded scene; Partially visible at image edge; Similar appearance (cap/hair/hood vs helmet)
10	hard_hat_workers447.jpg	no_helmet	helmet (0.38)	Wrong class	Similar appearance (cap/hair/hood vs helmet)
11	hard_hat_workers739.jpg	helmet	helmet (0.79)	Poor bounding box	Occlusion / crowded scene; Partially visible at image edge; Box boundary ambiguity / loose annotation
12	hard_hat_workers3684.jpg	no_helmet	no_helmet (0.59)	Poor bounding box	Partially visible at image edge; Box boundary ambiguity / loose annotation

Suggested improvements

- Add more training images for the weakest class person (test mAP@0.5:0.95 = 0.12) and rebalance classes with targeted collection.
- Small objects caused 98 errors: train at a higher resolution (imgsz=960) or use tiled/SAHI inference for far-away workers.
- Re-annotate the public images with missing labels (especially person) using the same guidelines as my manual batch - the audit showed the source labels are incomplete.
- Collect more low-light and motion-blurred site images and enable stronger brightness/blur augmentation.
- Add hard-negative examples of caps, hoods and headscarves labelled no_helmet to reduce helmet / no_helmet confusion.
- Train a larger model (YOLOv8m) for more epochs with cosine LR, and tune the confidence threshold per class to favour no_helmet recall.
- Test on real CCTV footage from local sites and add tracking (ByteTrack) so one violation is counted once per person, not once per frame.



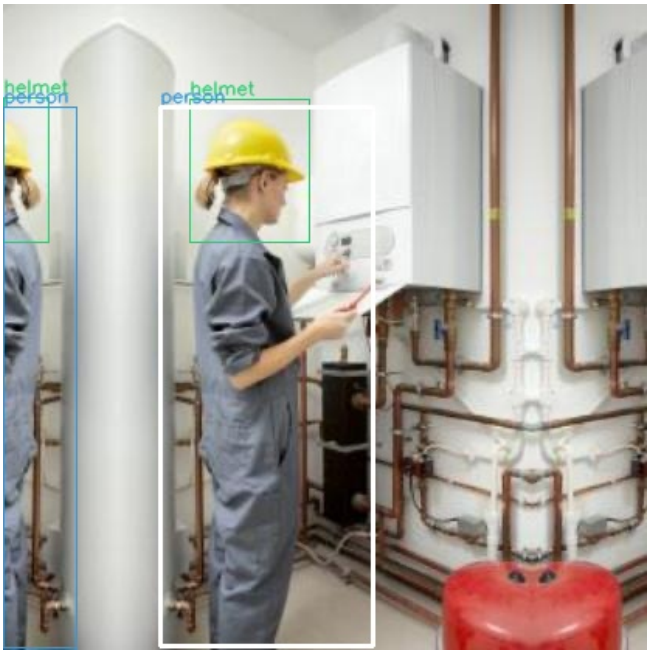
case 01 False negative



case 02 False negative



case 03 False negative



case 04 False negative

9. Demo

A Gradio web app (also deployable on Hugging Face Spaces) and a Streamlit app (app/streamlit_app.py) let a user upload an image and receive detections, per-class counts and the compliance status. Video inference results are saved in outputs/demo_results.



Frame from the annotated video output

10. LinkedIn-Ready Portfolio Report

1. Project Overview

I built a computer-vision safety system that detects whether workers are wearing safety helmets. Helmet rules are hard to enforce by watching cameras manually, so an automatic detector can flag violations as they happen. The model detects three classes - helmet, no_helmet (a bare head) and person - and for every image or video frame it draws labelled bounding boxes with confidence scores and marks the scene as COMPLIANT or VIOLATION. The goal was to go through the complete AI workflow, from data collection and annotation to training, evaluation, error analysis and a demo.

2. Tools and Technologies Used

Python, Google Colab (GPU: Tesla T4), Google Drive, YOLOv8, Ultralytics, OpenCV, Roboflow Annotate, Pandas, Matplotlib, ImageHash, Gradio and Streamlit.

3. Dataset Preparation

I started from the public Hard Hat Workers dataset (Kaggle, CC0 license, 5000 images), removed 92 near-duplicate images with perceptual hashing, and curated 500 images so that every class and difficult condition (small, dark, blurry and crowded scenes) is represented. Manual Roboflow batch of 120 images prepared (import pending), following strict annotation guidelines, and audited my labels against the original ones. I converted all annotations from Pascal VOC to YOLO format and created a stratified 70/20/10 split (349 train / 100 validation / 51 test).

4. Model Training

I trained a YOLOv8n baseline and a YOLOv8s experiment on Google Colab using pretrained COCO weights, image size 640 and 50 epochs (batch 16). The best model (yolov8n.pt) took 6.9 minutes on a Tesla T4 and was selected using validation mAP; its best checkpoint is saved as best.pt.

5. Evaluation Results

On the unseen test set the model reached precision 0.73, recall 0.65, mAP@0.5 0.68 and mAP@0.5:0.95 0.40. The best-performing class was no_helmet and the weakest was person. I also studied the confusion matrix and training/validation loss curves to understand behaviour across classes.

6. Inference Results

I tested the model on 51 unseen images: it draws bounding boxes, labels and confidence scores and reports a compliance status for each image. I also ran it on video (300 frames at about 55.8 FPS) and built a web demo where anyone can upload a photo.

7. Error Analysis

I matched every prediction with the ground truth and found 140 false negatives, 78 false positives, 6 wrong-class predictions and 7 poor boxes. The most common causes were small object, partially visible at image edge, occlusion / crowded scene. Some "false positives" were actually correct detections of people the original dataset never labelled - a reminder that label quality limits measured accuracy.

8. What I Learned

- How to choose, legally source and clean a computer-vision dataset (including duplicate removal)
- Manual bounding-box annotation and writing clear annotation guidelines
- Converting Pascal VOC annotations to the YOLO label format
- Training and comparing YOLOv8 models with transfer learning on Google Colab
- What precision, recall, mAP@0.5 and mAP@0.5:0.95 really measure
- That data quality and consistent labels matter as much as the model
- How systematic error analysis turns mistakes into a concrete improvement plan

9. Future Improvements

- Add more training images for the weakest class person (test mAP@0.5:0.95 = 0.12) and rebalance classes with targeted collection.
- Small objects caused 98 errors: train at a higher resolution (imgsz=960) or use tiled/SAHI inference for far-away workers.
- Re-annotate the public images with missing labels (especially person) using the same guidelines as my manual batch - the audit showed the source labels are incomplete.
- Collect more low-light and motion-blurred site images and enable stronger brightness/blur augmentation.
- Add hard-negative examples of caps, hoods and headscarves labelled no_helmet to reduce helmet / no_helmet confusion.

10. LinkedIn Post

Today, I completed an end-to-end Computer Vision project as part of the AIRI Team PITB AI Internship Task 1.

I built a Safety Helmet Detection System using YOLOv8 that flags workers without helmets in images and video.

The project covered the complete AI workflow:

- Problem selection
- Data collection (500 curated images, duplicates removed)
- Image annotation in Roboflow
- YOLO-format dataset preparation
- Model training on Google Colab (YOLOv8n vs YOLOv8s)

- Model evaluation (mAP@0.5 = 0.68 on unseen test images)
- Inference on unseen images and video
- Error analysis
- Final documentation

This project helped me understand how real-world Computer Vision systems are developed beyond just training a model. I learned the importance of dataset quality, annotation consistency, evaluation metrics, and error analysis.

Tools used: Python, Google Colab, YOLOv8, Ultralytics, OpenCV, and Roboflow.

Key learning: A good AI model depends not only on the algorithm, but also on clean data, correct labels, proper evaluation, and continuous improvement.

#ComputerVision #DeepLearning #YOLOv8 #ArtificialIntelligence #MachineLearning #Python #AIRI #PITB #AIInternship

References

- Hard Hat Workers dataset, MakeML - Kaggle: andrewmvd/hard-hat-detection (CC0)
- Ultralytics YOLOv8 - https://docs.ultralytics.com
- Roboflow Annotate - https://roboflow.com